

Online Event Management System – Project Explanation

This project is a **database-driven Online Event Management System** designed to help users **register, browse, and book events online**, while allowing admins to **manage events, users, discounts, and payments** efficiently.

Problem it Solves

Managing events manually is time-consuming and error-prone—especially when handling:

- Multiple event types
- User registrations
- Discounts and special preferences
- Payments and bookings

This system provides a **centralized, structured database** that automates all these processes.

Key Actors

1. User

- Registers using email and password
- Browses events (Dance, Music, Arts, Sports, etc.)
- Books events with number of seats
- Applies **discounts** and **special preferences**
- Makes payment using multiple modes
- Can update or cancel bookings

2. Admin

- Logs in with admin credentials
- Adds, updates, or deletes:
 - Events
 - Discounts
 - Special preferences
 - User details
- Manages pricing and discount rates

Core Database Modules (Entities)

User

Stores registered user details:

- User ID, Email, Username, Password
- Users **must register before booking** any event.

Event Details

Stores complete event information:

- Event name, type, date & time
 - Duration and base price
- Each event belongs to a specific **event category**.

Event Discount

Manages discount options:

- Premium account
- Group booking
- Coupon-based discounts

Each discount has a **rate reduced** value.

Special Preference

Optional add-ons for users:

- VIP seats
- Backstage access
- Exclusive services

These come with **extra charges**.

User Booking

Acts as the **core linking table**:

- Connects User, Event, Discount, and Special Preference
- Stores number of seats booked
- Each booking has a unique **Booking ID (BID)**

This table enforces **referential integrity using foreign keys**.

Amount (Payment)

Handles payment details:

- Payment ID
- Booking ID
- Mode of payment (Credit, Debit, Wallet, etc.)

Admin

Stores admin login credentials and permissions.

Database Design Highlights

- Fully **normalized (3NF)** to avoid redundancy
- Strong **foreign key relationships**
- Uses **indexes** for faster search
- Supports **views** for business insights

Important SQL Features Used

Queries

- Simple queries (filter, sort)
- Aggregate queries (COUNT, SUM, AVG)
- Join queries (INNER, LEFT, RIGHT joins)

Views

Examples:

- Total amount payable by each user (including discounts & extra charges)
- Users who registered but never booked
- Number of bookings per user
- Events sorted by price for easy searching

Indexes

- Faster event search by type & name
- Unique email constraint for users
- Faster lookup for bookings, discounts, and prices

Overall Summary (One-Line)

This project implements a **relational database-based Online Event Management System** that efficiently handles user registrations, event bookings, discounts, special preferences, and payments with strong data integrity and optimized query performance.